

FLI1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FLI1 gene fusions from the msk_impact_50k_2026 study, covering 118 fusion events. 73/118 fusions (61.9%) were in-frame. 7/118 fusions (5.9%) retained the Sterile alpha motif (SAM)/Pointed domain. Domain retention was tested with Fisher's exact test ($p=0.988837$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FLI1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 118 analyzed fusion events, identifying 3 distinct fusion partner genes. The most frequent partner was EWSR1, observed in 116 events.

Domain retention

Sterile alpha motif (SAM)/Pointed domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/73 (2.7%) of in-frame fusions retained the domain, $p=0.988837$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.80198.

Domain disruption

Domain-disruption analysis was skipped: FLI1 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 117 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 241 aa (permutation-corrected $p=0.019802$, statistically significant at $\alpha=0.05$). This is -39 aa from the nearest configured domain boundary at 280 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared `Domain_retention_flags` groups "retained" ($n=7$) and "not_retained" ($n=110$). For `Frame_status`, retained: 2/5 (40.0%, 95% CI 11.8%-76.9%); not_retained: 71/102 (69.6%, 95% CI 60.1%-77.7%). Welch's t-test comparing `Tumor_variant_count` between groups gave means of 20.7143 vs 66.9091, $p=2.40299e-05$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: FLI1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EWSR1	116	0.9830508474576272	0.00958363549039928	None	0.7245283018867925	0.5862586106414931	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.5807703368213202	1
ATM	1	0.00847457627118644	0.00958363549039928	None	0.8113207547169812	0.5862586106414931	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.23157782003206526	2
ERG	1	0.00847457627118644	0.00958363549039928	None	0.0	0.5862586106414931	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.07945517852263131	3

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
129	0	7	3	107	0.0	1.0	-0.0
197	1	6	29	81	0.46551724137931033	0.6759093008029847	0.1701115774363377
219	4	3	105	5	0.06349206349206349	0.0065847756701910435	2.1814590159729152
241	4	3	107	3	0.037383177570093455	0.00248300753782759	2.6050219620274095
261	6	1	109	1	0.05504587155963303	0.11656351311523727	0.9334373718114223
277	6	1	110	0	0.0	0.059829059829059825	1.2230878217319048

domain_retention tables

frame_domain_contingency_table

2	5
71	39

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Sterile alpha motif (SAM)/Pointed domain	117	7	29	81	0.02552026286960464

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
3	7	118	0.059322033898305086

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ATM	1
ERG	1
EWSR1	116

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024080-T02-IM6-1	P-0024080-T02-IM6		ATM-FLI1	ATM	out-of-frame	five_prime	128677281	7	261	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			FLI1 (NM_002017) - ATM (NM_000051) rearrangement: c.781+147:FLI1_c.8269-128:ATMdup Protein Fusion: out of frame {FLI1:ATM}	NA
EVT-P-0000253-T02-IM5-2	P-0000253-T02-IM5		EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128664485	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1(NM_013986)-FLI1(NM_002017) Fusion (EWSR1 exon 9 fused to FLI1 6) : t(22;11)(22q12.2.13;11q24.3)(chr22:g.29685870::chr11:g.128664485) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0047480-T03-I-M6-3	P-0047480-T03-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128649430	5	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128649430::chr22:g.29688384) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0001377-T01-I-M3-4	P-0001377-T01-IM3		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128660474	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		None Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0001896-T01-I-M3-5	P-0001896-T01-IM3		EW-SR1-FLI1	EW-SR1	unknown	three_prime	128668840	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) Translocation: t(22;10)(22q12.2;10q26.2)(chr22:g.29684628::chr10:g.128668840) Protein fusion: mid-exon (EWSR1-FLI1)	NA
EVT-P-0002240-T01-I-M3-6	P-0002240-T01-IM3		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128638556	3	129	True	disrupted	0.8313253012048193	True			Sterile alpha motif (SAM)/Pointed domain	NA Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0003272-T01-I-M5-7	P-0003272-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128673862	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		NA Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0003290-T01-I-M5-8	P-0003290-T01-IM5		EW-SR1-FLI1	EW-SR1	unknown	three_prime	128641490	4	129	True	disrupted	0.8313253012048193	True			Sterile alpha motif (SAM)/Pointed domain	NA Protein fusion: mid-exon (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0003554-T01-IM5-9	P-0003554-T01-IM5		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128664769	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exon 9 fused to FLI1 exon 6): t(22;11)(q12.2;q24.3)(chr22:g.29685775::chr11:g.128664769) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0004498-T01-IM5-10	P-0004498-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128674058	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) reciprocal fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.3)(chr11:g.128674058::chr22:g.29688226) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0004931-T02-IM5-11	P-0004931-T02-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128651438	5	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) reciprocal fusion (EWSR1 exons 1-10 with FLI1 exons 5-9) : t(22;11)(22q12.2;11q24.3)(chr22:g.29688357::chr11:g.128651438) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0005297-T01-IM5-12	P-0005297-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128658432	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 with FLI1 exons 6-9): t(22;11)(22q12.2;11q24.3)(chr22:g.29684524::chr11:g.128658432) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0005297-T02-IM6-13	P-0005297-T02-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128658433	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 with FLI1 exons 6-9): t(22;11)(22q12.2;11q24.3)(chr22:g.29684524::chr11:g.128658433) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0005614-T01-IM5-14	P-0005614-T01-IM5		EW-SR1-FLI1	EW-SR1	out-of-frame	five_prime	128667277	6	219	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.2)(chr11:g.128667277::chr22:g.29684957) Protein fusion: out of frame (FLI1-EWSR1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0005716-T01-I M5-15	P-0005716-T01-IM5		EW SR1-FLI1	EW SR1	in-frame	three_prime	128662783	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(22;11)(q12.2;q12.3)(chr22:g.29688428::chr11:g.128662783) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0005977-T01-I M5-16	P-0005977-T01-IM5		EW SR1-FLI1	EW SR1	out-of-frame	three_prime	128658321	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(22;11)(q12.2;q12.3)(chr22:g.29685512::chr11:g.128658321) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0006040-T01-I M5-17	P-0006040-T01-IM5		EW SR1-FLI1	EW SR1	out-of-frame	three_prime	128656157	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(22;11)(q12.2;q12.3)(chr22:g.29685318::chr11:g.128656157) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0008030-T01-I M5-18	P-0008030-T01-IM5		EW SR1-FLI1	EW SR1	out-of-frame	three_prime	128647136	4	197	True	disrupted	0.012048192771084338	True		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 5-9): t(11;22)(11q24.3;q12.2)(chr11:g.128647136::chr22:g.29685491) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0008102-T01-I M5-19	P-0008102-T01-IM5		EW SR1-FLI1	EW SR1	in-frame	three_prime	128649909	5	197	True	disrupted	0.012048192771084338	True		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_001163287) - FLI1 (NM_002017) fusion (EWSR1 exons 1-7 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128649909::chr22:g.29684432) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0008320-T01-I M5-20	P-0008320-T01-IM5		EW SR1-FLI1	EW SR1	in-frame	three_prime	128651930	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128651930::chr22:g.29683853) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0008352-T01-IM5-21	P-0008352-T01-IM5		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128658456	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-7 with FLI1 exons 6-9) : t(22;11)(22q12.2;11q24.3)(chr22:g.29685587::chr11:g.128658456) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0008355-T01-IM5-22	P-0008355-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128673264	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-7 fused with FLI1 exons 6-9) : t(22;11)(22q12.2;11q24.3)(chr22:g.29683339::chr11:g.128673264) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0008731-T01-IM5-23	P-0008731-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128647421	5	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9): t(11;22)(11q24.3;22q12.2)(chr11:g.128647421::chr22:g.29684579) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0012131-T01-IM5-25	P-0012131-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128663970	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion : t(11;22)(q24.3;q12.2)(chr11:g.128663970::chr22:g.29683265) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0012316-T01-IM5-26	P-0012316-T01-IM5		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128674054	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) -FLI1(NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128674054::chr22:g.29688203) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0012482-T02-IM5-27	P-0012482-T02-IM5		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128639397	3	129	True	disrupted	0.8313253012048193	True		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 4-9): t(11;22)(q24.3;q12.2)(chr11:g.128639397::chr22:g.29684794) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 3601-T01-I M5-28	P-0013 601-T0 1-IM5		EW SR1 -FLI 1	EW SR1	out- of-fr ame	thr ee _pr im e	128664 219	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128664219::chr22:g.29685257) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-001 3746-T02-I M5-29	P-0013 746-T0 2-IM5		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee _pr im e	128644 899	4	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused to FLI1 exons 5 - 9): t(11;22)(q24.3;q12.2)(chr11:g.28644899::chr22:g.29688263) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-001 3746-T04-I M7-30	P-0013 746-T0 4-IM7		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee _pr im e	128644 900	4	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644900::chr22:g.29688264) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 3874-T01-I M5-31	P-0013 874-T0 1-IM5		EW SR1 -FLI 1	EW SR1	in-fr ame	five _pr im e	128677 168	7	261	True	retai ned	1.0	False	Sterile alpha motif (S AM)/ Pointe d dom ain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 with FLI1 exons 8-9) : t(11;22)(q24.3;q12.2)(chr11:g.128677168::chr22:g.29688431) Protein fusion: in frame (FLI1-EWSR1)	NA
EVT-P-006 5945-T03-I M7-32	P-0065 945-T0 3-IM7		EW SR1 -FLI 1	EW SR1	out- of-fr ame	thr ee _pr im e	128656 142	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128656142::chr22:g.29684998) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-001 4314-T01-I M6-33	P-0014 314-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee _pr im e	128663 756	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128663756::chr22:g.29684463) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 4984-T02-I M6-34	P-0014 984-T0 2-IM6		EW SR1 -FLI 1	EW SR1	out- of-fr ame	three _prim e	128660 148	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128660148::chr22:g.29684827) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-001 5517-T01-I M6-36	P-0015 517-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out- of-fr ame	three _prim e	128653 928	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.2)(chr11:g.29685072::chr22:g.29685072) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-001 5619-T03-I M6-37	P-0015 619-T0 3-IM6		EW SR1 -FLI 1	EW SR1	out- of-fr ame	three _prim e	128663 112	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128663112::chr22:g.29685920) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-001 6346-T01-I M6-38	P-0016 346-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	three _prim e	128651 008	5	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128651008::chr22:g.29683915) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 7027-T01-I M6-39	P-0017 027-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	three _prim e	128646 363	4	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused in-frame with FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.128646363::chr22:g.29684426) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 8508-T01-I M6-42	P-0018 508-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	three _prim e	128653 753	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused in-frame with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128653753::chr22:g.29684479) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 8530-T01-I M6-43	P-0018 530-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128643 225	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128643225::chr22:g.29684272) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 8663-T01-I M6-44	P-0018 663-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128658 406	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128658406::chr22:g.29684132) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 9080-T02-I M6-45	P-0019 080-T0 2-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128663 272	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128663272::chr22:g.29683811) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 9756-T01-I M6-46	P-0019 756-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128678 982	8	261	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 8-9) : t(11;22)(q24.3;q12.2)(chr11:g.128678982::chr22:g.29683784) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-002 0463-T01-I M6-47	P-0020 463-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128668 415	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128668415::chr22:g.29683880) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 1068-T01-I M6-48	P-0021 068-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128660 877	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128660877::chr22:g.29683596) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-002 1939-T01-I M6-49	P-0021 939-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128665 058	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exon 8 fused to FLI1 exon 6): t(11;22)(q24.3;q12.2)(chr11:g.128665058::chr22:g.29683347) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-002 2452-T01-I M6-50	P-0022 452-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128672 031	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1- 11 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128672031::chr22:g.29688410) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-000 0719-T01-I M3-51	P-0000 719-T0 1-IM3		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128673 072	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		NA Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-002 4807-T01-I M6-52	P-0024 807-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128656 386	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128656386::chr22:g.29684194) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 4945-T01-I M6-53	P-0024 945-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128663 120	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128663120::chr22:g.29683825) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-002 5251-T01-I M6-54	P-0025 251-T0 1-IM6		EW SR1 -FLI 1	EW SR1	unknown	five_prime	128662 338	5	219	True	retained	1.0	False	Sterile alpha motif (S AM)/ Pointed domain			FLI1 (NM_002017) - EWSR1 (NM_013986) Rearrangement : t(11;22)(q24.3;q12.2)(chr11:g.128662338::chr22:g.29683006) Protein Fusion: mid-exon {FLI1:EWSR1}	NA
EVT-P-002 6404-T01-I M6-55	P-0026 404-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128652 868	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr22:g.29684987::chr11:g.128652868) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-002 9448-T01-I M6-56	P-0029 448-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128647 971	5	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128647971::chr22:g.29684099) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-002 9619-T02-I M6-57	P-0029 619-T0 2-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128662 958	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662958::chr22:g.29688349) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 0044-T02-I M6-58	P-0030 044-T0 2-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128653 122	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_005243) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128653122::chr22:g.29684176) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sampl e_id	pa tie nt _i d	fusi on_ na me	part ner_ _ge ne	fra me_ _sta tus	tar get_ _ro le	breakp oint_ge nomic	break point_ _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ _statu s	domain_r etained_fr action	domain_ _is_trun cated	retain ed_do mains	lost_ _do mains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 0044-T03-I M7-59	P-0030 044-T0 3-IM7		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee_ _pr im e	128653 122	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128653122 ::chr22:g.29684176) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 0220-T01-I M6-60	P-0030 220-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee_ _pr im e	128676 517	7	241	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128676517 ::chr22:g.29688327) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 0279-T02-I M6-61	P-0030 279-T0 2-IM6		EW SR1 -FLI 1	EW SR1	unk now n	thr ee_ _pr im e	128662 512	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128662512 ::chr22:g.29684687) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-003 0279-T03-I M7-62	P-0030 279-T0 3-IM7		EW SR1 -FLI 1	EW SR1	unk now n	thr ee_ _pr im e	128662 512	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128662512 ::chr22:g.29684687) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-003 0349-T03-I M6-63	P-0030 349-T0 3-IM6		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee_ _pr im e	128666 334	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) -FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q1 2.2)(chr11:g.128666334::chr22: g.29683948 Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2382-T01-I M6-64	P-0032 382-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128676 273	7	241	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused in-frame with FLI1 exons 7-9): t(11;22)(q24.3;q12.2)(chr11:g.128676273::chr22:g.29688342) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 2610-T01-I M6-65	P-0032 610-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128644 868	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644868::chr22:g.29685790) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-003 4612-T01-I M6-66	P-0034 612-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128659 530	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128659530::chr22:g.29683806) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 4612-T02-I M6-67	P-0034 612-T0 2-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128659 530	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128659530::chr22:g.29683806) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 6245-T01-I M6-68	P-0036 245-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128658 093	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128658093::chr22:g.29684449) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 7173-T01-I M6-69	P-0037 173-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128670 710	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670710::chr22:g.29688297) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 7267-T01-I M6-70	P-0037 267-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128672 985	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128672985::chr22:g.29688402) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 8127-T01-I M6-71	P-0038 127-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128646 529	4	197	True	disrupted	0.0120481 92771084 338	True		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.28646529::chr22:g.29683923) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 8570-T02-I M6-72	P-0038 570-T0 2-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128664 954	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128664954::chr22:g.29683885) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 8570-T03-I M6-73	P-0038 570-T0 3-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128664 954	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128664954::chr22:g.29683885) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 9202-T02-I M7-74	P-0039 202-T0 2-IM7		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128670 148	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670148::chr22:g.29688434) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 9406-T01-I M6-75	P-0039 406-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128652 054	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) rearrangement: t(11;22)(q24.3;q12.2)(chr11:g.128652054::chr22:g.29685781) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-003 9579-T01-I M6-76	P-0039 579-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128665 386	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) rearrangement: t(11;22)(q24.3;q12.2)(chr11:g.128665386::chr22:g.29684257) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-003 9688-T01-I M6-77	P-0039 688-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128678 290	8	261	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128678290::chr22:g.29684082) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 0442-T01-I M6-78	P-0040 442-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128673 571	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673571::chr22:g.29685161) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 0442-T02-I M6-79	P-0040 442-T0 2-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128673 571	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673571::chr22:g.29685161) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-004 1943-T01-I M6-80	P-0041 943-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128644 404	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644404::chr22:g.29683564) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 2913-T01-I M6-81	P-0042 913-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128664 632	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128664632::chr22:g.29684449) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 3526-T01-I M6-82	P-0043 526-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128660 704	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660704::chr22:g.29684397) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 3526-T03-I M6-83	P-0043 526-T0 3-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128660 704	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660704::chr22:g.29684397) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 4551-T01-I M6-84	P-0044 551-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128655 556	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128655556::chr22:g.29683586) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 4896-T02-I M6-85	P-0044 896-T0 2-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128658 358	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128658358::chr22:g.29685203) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-004 5227-T01-I M6-86	P-0045 227-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128647 321	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684323) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 5227-T03-I M6-87	P-0045 227-T0 3-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128647 321	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684323) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 5722-T01-I M6-88	P-0045 722-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128679 847	9	277	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128679847::chr22:g.29683765) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 6693-T01-I M6-89	P-0046 693-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128656 732	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128656732::chr22:g.29683940) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 7109-T02-IM6-90	P-0047 109-T02-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128666244	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128666244::chr22:g.29683718) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-001 3922-T01-IM5-91	P-0013 922-T01-IM5		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128644690	4	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-10 fused to FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.128644690::chr22:g.29688612) Protein fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-004 8417-T02-IM6-92	P-0048 417-T02-IM6		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128662931	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662931::chr22:g.29685071) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-006 8585-T02-IM7-93	P-0068 585-T02-IM7		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128672830	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128672830::chr22:g.29683515) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-004 8577-T03-IM6-94	P-0048 577-T03-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128669947	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128669947::chr22:g.29683727) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 1333-T01-I M6-95	P-0051 333-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128667 777	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128667777::chr22:g.29684190) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-005 1569-T01-I M6-96	P-0051 569-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128643 608	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128643608::chr22:g.29684917) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-005 2277-T01-I M6-97	P-0052 277-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128649 484	5	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128649484::chr22:g.29685235) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-005 2635-T01-I M6-98	P-0052 635-T0 1-IM6		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128660 258	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660258::chr22:g.29685095) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-005 2745-T02-I M6-99	P-0052 745-T0 2-IM6		EW SR1 -FLI 1	EW SR1	unknown	three_prime	128670 137	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670137::chr22:g.29688551) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-006 6379-T02-I M7-100	P-0066 379-T0 2-IM7		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128643 844	4	197	True	disrupted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128643844::chr22:g.29691291) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0053879-T02-I M6-101	P-0053879-T02-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128665052	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128665052::chr22:g.29688409) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0053991-T01-I M6-102	P-0053991-T01-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128647321	4	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684445) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0055969-T02-I M6-103	P-0055969-T02-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128674720	6	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128674720::chr22:g.29683823) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0056323-T02-I M6-104	P-0056323-T02-IM6		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128659413	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128659413::chr22:g.29685696) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0056509-T02-I M6-105	P-0056509-T02-IM6		EW-SR1-FLI1	EW-SR1	out-of-frame	three_prime	128648407	5	197	True	disrupted	0.012048192771084338	True			Sterile alpha motif (SAM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128648407::chr22:g.29685275) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0056693-T02-I M6-106	P-0056693-T02-IM6		EW-SR1-FLI1	EW-SR1	in-frame	three_prime	128654385	5	219	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128654385::chr22:g.29683723) Protein Fusion: in frame {EWSR1:FLI1}	NA

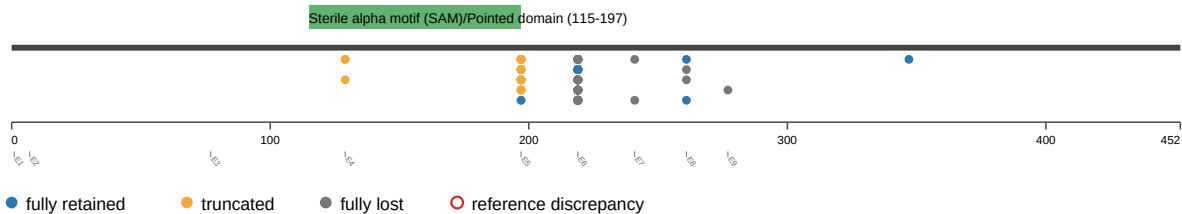
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 6712-T01-I M6-107	P-0056 712-T0 1-IM6		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128672 121	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128672121::chr22:g.29684489) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-005 7669-T01-I M6-108	P-0057 669-T0 1-IM6		EW SR1 -FLI 1	EW SR1	unknown	three_prime	128673 876	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673876::chr22:g.29688565) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-005 9143-T01-I M7-109	P-0059 143-T0 1-IM7		EW SR1 -FLI 1	EW SR1	unknown	three_prime	128660 670	5	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660670::chr22:g.29684654) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-005 9518-T02-I M7-110	P-0059 518-T0 2-IM7		EW SR1 -FLI 1	EW SR1	in-frame	three_prime	128671 917	6	219	True	lost	0.0	False		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128671917::chr22:g.29684466) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-006 0347-T02-I M7-111	P-0060 347-T0 2-IM7		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128648 014	5	197	True	disrupted	0.0120481 92771084 338	True		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128648014::chr22:g.29685134) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-006 1850-T02-I M7-112	P-0061 850-T0 2-IM7		EW SR1 -FLI 1	EW SR1	out-of-frame	three_prime	128646 072	4	197	True	disrupted	0.0120481 92771084 338	True		Sterile alpha motif (S AM)/ Pointed domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128646072::chr22:g.29685975) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sampl_e_id	pa_tie_nt_i_d	fusi_on_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_d_o_mains	lost_d_o_mains	disrupt_ed_d_o_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 2142-T03-I M7-113	P-0062 142-T0 3-IM7		EW SR1 -FLI 1	EW SR1	out- of-fr ame	thr ee _pr im e	128644 385	4	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128644385 ::chr22:g.29685696) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-006 2142-T04-I M7-114	P-0062 142-T0 4-IM7		EW SR1 -FLI 1	EW SR1	out- of-fr ame	thr ee _pr im e	128644 385	4	197	True	disru pted	0.0120481 92771084 338	True			Sterile alpha motif (S AM)/ Pointed domain	EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128644385 ::chr22:g.29685696) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-006 3940-T02-I M7-115	P-0063 940-T0 2-IM7		EW SR1 -FLI 1	EW SR1	unk now n	thr ee _pr im e	128672 287	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128672287 ::chr22:g.29688521) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-006 5014-T01-I M7-116	P-0065 014-T0 1-IM7		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee _pr im e	128666 909	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128666909 ::chr22:g.29684574) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-006 5014-T02-I M7-117	P-0065 014-T0 2-IM7		EW SR1 -FLI 1	EW SR1	in-fr ame	thr ee _pr im e	128666 909	6	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128666909 ::chr22:g.29684574) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-006 5672-T01-I M7-118	P-0065 672-T0 1-IM7		EW SR1 -FLI 1	EW SR1	out- of-fr ame	thr ee _pr im e	128658 571	5	219	True	lost	0.0	False		Steri le al pha moti f (S AM)/ Poin ted dom ain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q 24.3;q12.2)(chr11:g.128658571 ::chr22:g.29685397) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001233-T01-IM3-119	P-0001233-T01-IM3		EWSR1-FLI1	EWSR1	out-of-frame	five_prime	128646455	4	197	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			EWSR1 (NM_005243) - FLI1 (NM_002017) Translocation: t(22;11)(q12.2;q24.3)(c.974+158_c.589+3575) Protein fusion: out of frame (FLI1-EWSR1)	NA
EVT-P-0053737-T01-IM6-120	P-0053737-T01-IM6		FLI1-ERG	ERG	unknown	five_prime	128680564	9	347	False	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			FLI1 (NM_002017) - ERG (NM_182918) rearrangement: t(11;21)(q24.3;q22.2)(chr11:g.128680564::chr21:g.39755655) Protein Fusion: mid-exon {FLI1:ERG}	NA
EVT-P-0050027-T02-IM6-121	P-0050027-T02-IM6		FLI1-EWSR1	EWSR1	in-frame	five_prime	128669484	6	219	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			FLI1 (NM_002017) - EWSR1 (NM_013986) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128669484::chr22:g.29683917) Protein Fusion: in frame {FLI1:EWSR1}	NA

domain retention outliers

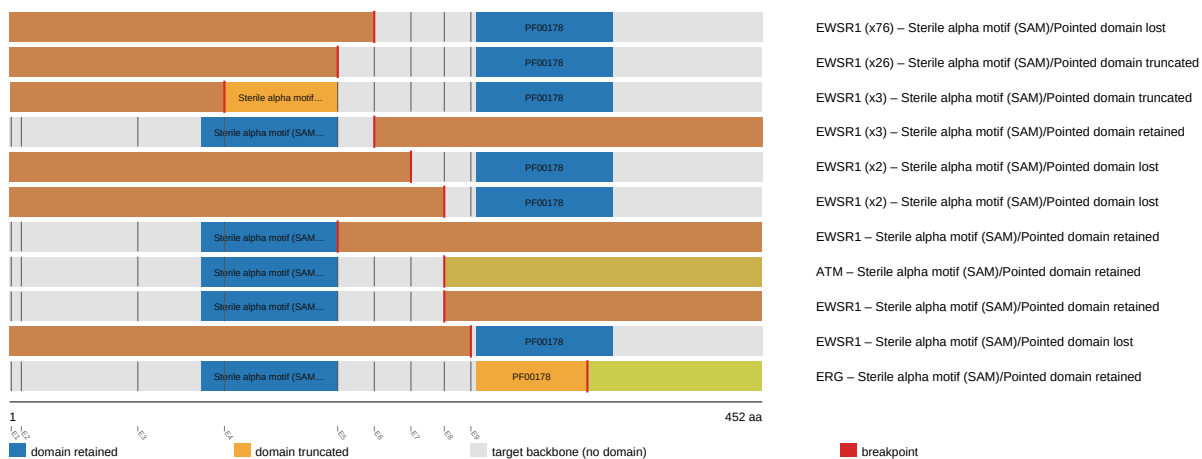
FLI1 domain-retention track



fusion schematic

FLI1 fusion-transcript schematic

partner block → breakpoint → retained FLI1 portion (domain-colored, exon ticks)



reference comparison

Reference vs msk_impact_50k_2026

